

The arc is a good instrument; the map it draws is the room. Below 3 kHz the hub loudspeaker is a valid axisymmetric source, and from 3–5 kHz it is not.

Change one thing at a time

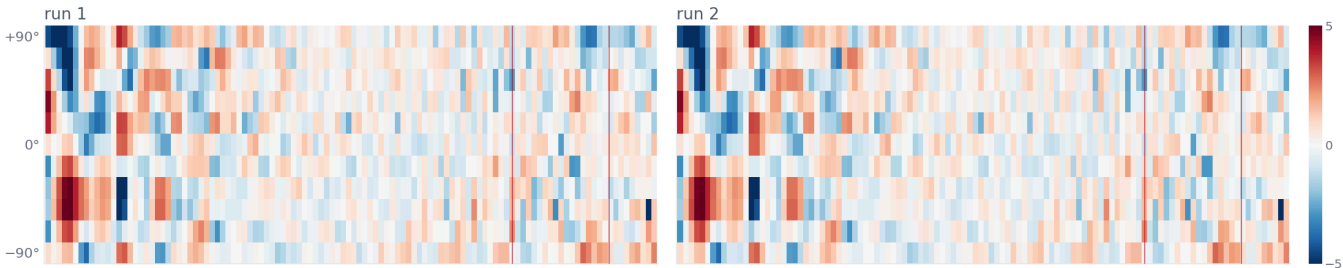
Eleven UMIK-2 capsules on a 1.68 m arc · 112 tones, 257–6350 Hz · each cell is one capsule at one tone, dB re the eleven-capsule mean per-capsule calibration applied · red rules mark 3–5 kHz, where the source is not axisymmetric

1 Nothing changed

the same capture twice, 20 min apart

0.03 dB

r = +1.00



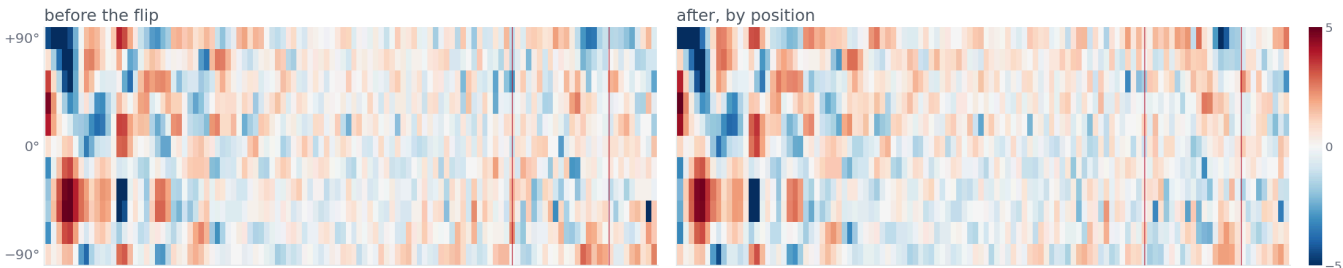
the instrument is silent when nothing moves

2 The arc flipped 180°

second run re-indexed to where each capsule physically sat

0.53 dB

r = +0.92



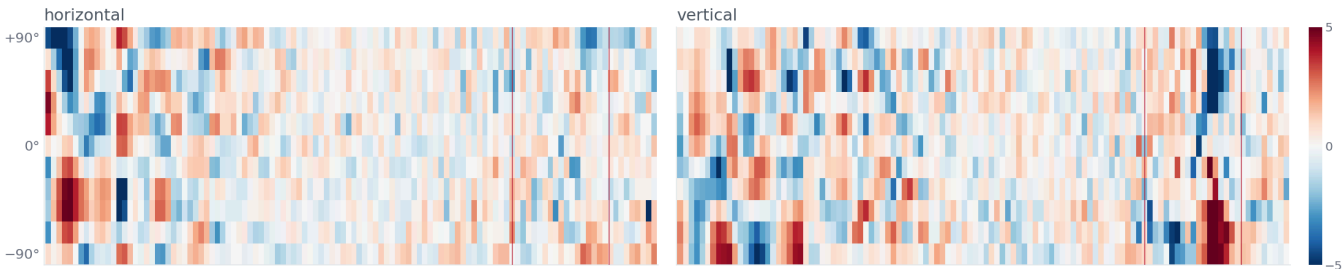
the pattern stayed with the room, not with the arc

3 The arc stood vertical

same eleven capsules, clamps and cables — a different room

2.29 dB

r = -0.12



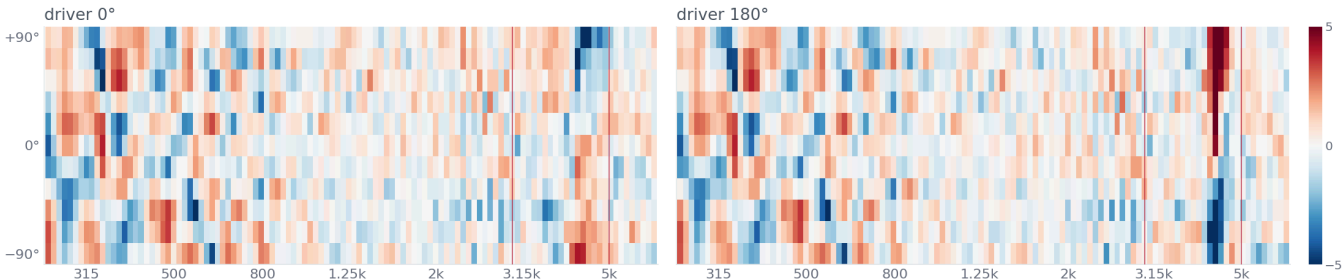
change the room and the map is unrecognisable

4 Only the driver turned 180°

inside a sphere that never moved, arc vertical both times

0.20 dB

below 3 kHz



and 11.0 dB at 4362 Hz — the source's own rocking mode

0.21 dB

of the position map is provably the rig — antisymmetric under the flip

1.04 dB

is provably the room; the 0.84 dB a single flip cannot divide is room too (cov = -0.09 dB², CI [-0.22, +0.03])

≤ 0.16 dB

is all the two geometries share — so the rig's total contribution is ~0.2 dB

11.0 dB

at 4362 Hz when only the driver turns: an m = 1 rocking mode, ka = 2.39. Avoid 3–5 kHz

Open. The loudspeaker map does not reproduce the propeller map (r = -0.04 tone-notched, +0.04 mixed; all 22 relabellings tested, best +0.24) — the propeller's own inflow, or the object at the hub, still matters. The absolute anchor needs a 94 dB calibrator. Room geometry is unrecorded: hub height, room L×W×H, foam, what stands near the arc ends.

Data. `calibrator/sessions/2026-09-16` (capsules) and `2026-09-17` (arc); levels and meta mirrored to the measurement repo, WAVs local. `calibrator/rig.py` captures and `read_map()` is the single loader, so a figure cannot disagree with the capture about what a level means; the bad band is `SOURCE_SUSPECT_HZ`. Every number here was recomputed from those captures while this page was written. Where this page and `CLAUDE.md` disagree, `CLAUDE.md` governs. rev 2, 2026-09-17.